

APIx

Real-time Airfare Price Index — infrastructure-grade fare tracking for India's domestic air routes



PROBLEM STATEMENT ID

SIH26056

CATEGORY

Software

THEME

[Add Theme]

TEAM NAME

[Add Team Name]

IDEA / PROPOSED SOLUTION

A defensible, transparent airfare index — built for policy, not for deal-hunting

THE PROBLEM

India has no real-time, methodology-transparent airfare index. OTA fare trackers are commercial, opaque, and unfit for use by statistical or monetary authorities like the NSO or RBI as an inflation input.

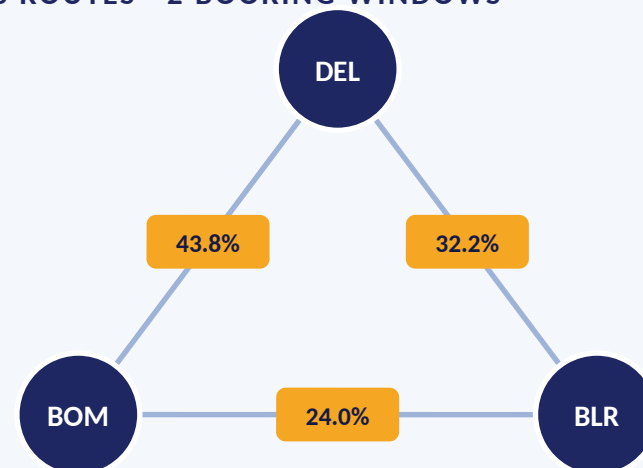
THE SOLUTION — APIx

One defensible index — DGCA-traffic-weighted, chain-linked, the same logic behind CPI & the GDP deflator.

Two independent real sources (Air India + IndiGo direct) plus a clearly-labelled synthetic gap-filler — never blended silently.

Fully traceable — every number on every page maps to a real database-backed API call.

3 ROUTES • 2 BOOKING WINDOWS



Weights = DGCA Annual Traffic Survey share, FY22-23

TECHNICAL APPROACH

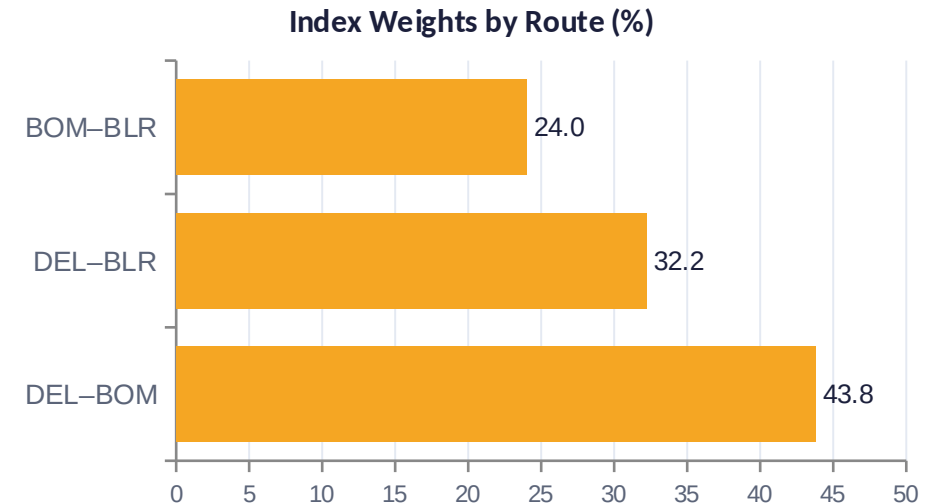
From scrape to screen — a five-stage, fully traceable pipeline



THE FORMULA

$$I(t) = I(t-1) \cdot \exp\left(\sum w(\text{route}) \cdot \ln\left(\frac{\text{fare}(\text{route}, t)}{\text{fare}(\text{route}, t-1)}\right)\right)$$

Traffic-weighted like CPI food weighting · chain-linked like India's GDP deflator



FEASIBILITY AND VIABILITY

Built to survive judge scrutiny — every risk named and closed



Single-source bias

Two independent airline-direct sources (Air India + IndiGo), pluggable connector interface for more.



Synthetic data dominating

Gap-filler only — always tagged `synthetic\_estimate`, never blended silently with real quotes.



Arbitrary index weights

Weights sourced directly from DGCA's Annual Traffic Survey, shown on-screen with citation.



Scraping legality

robots.txt reviewed for all 5 candidate sites; every OTA (Ixigo, EaseMyTrip, Cleartrip) disallows bots — excluded. Airline-direct sites, confirmed permissive, used instead.

Zero-cost stack: Supabase (free Postgres) · Render/Railway (backend) · Vercel/Netlify (frontend) — no infra cost to run or judge.

IMPACT AND BENEFITS

Policy-grade infrastructure, not another fare-deal app

NSO

Enriches the CPI travel-cost basket with a transparent, auditable airfare series.

RBI

An early-warning signal for travel-cost inflation feeding monetary policy analysis.

DGCA / Policy makers

Independent cross-check against carrier-reported fares, route by route.

Researchers

Open, documented API and methodology — reproducible, not a black box.

3

Routes tracked

2

Independent sources

8

Dashboard pages

\$0

Infra cost

Grounded in official data and established index methodology

Methodology precedent

- IMF Manual on Consumer Price Indices, Ch. 15 — chain-linking
- India GDP Deflator methodology — chain-linking precedent
- DGCA Annual Traffic Survey, FY22-23 — route traffic weights (dgca.gov.in)

Data & compliance

- airindia.com/robots.txt & goindigo.in/robots.txt — reviewed 2026-08-24
- OTA robots.txt review — Ixigo, EaseMyTrip, Cleartrip (all disallow bots; excluded)
- Polite-scraping protocol — rate limits, capped volume, non-impersonating UA

Technology

- FastAPI, Supabase Postgres — backend & storage
- React 19 + TypeScript + Vite + Tailwind + Recharts — frontend
- Playwright, Pandas/NumPy — ingestion & cleaning